Student Grade Analyzer Toolkit Documentation

1. Title & Objective

Title:

"Building a Student Grade Analyzer Using NumPy – A Beginner's Guide"

Objective:

This project demonstrates how to use **NumPy** in Python to analyze student performance data by calculating statistical measures and assigning grades automatically.

Technology Chosen:

- **NumPy (Python library)**

Why NumPy?

- Efficient for handling numerical data
- Provides built-in statistical functions
- Widely used in data science and machine learning

End Goal:

To build a program that:

- Stores student scores
- Calculates statistics (mean, median, standard deviation)
- Assigns grades (A–F)
- Displays results in a clear format

2. Quick Summary of the Technology

What is NumPy?

NumPy (Numerical Python) is a Python library used for working with arrays and performing fast mathematical operations.

Where is it used?

- Data science
- Machine learning
- Scientific computing
- Financial analysis

Real-world example:

Companies like Netflix use data analysis tools (including NumPy-based workflows) to analyze user behavior and improve recommendations.

3. System Requirements

- **Operating System:** Windows / Mac / Linux
- **Editor:** Visual Studio Code
- **Programming Language:** Python 3.x
- **Libraries:**
  - NumPy

4. Installation & Setup Instructions

Step 1: Install Python

Download from: https://www.python.org/downloads/

Step 2: Install NumPy

```
pip install numpy
```

OR (if using Anaconda):

```
conda install numpy
```

Step 3: Create Project Folder

```
mkdir student-grade-analyzer
cd student-grade-analyzer
```

Step 4: Create Files

```
main.py
README.md
requirements.txt
```

Step 5: Run the Program

```
python main.py
```

5. Minimal Working Example

Description:

This program stores student scores, calculates statistics, and assigns grades.

Code:

```
import numpy as np

# Store scores
scores = np.array([78, 85, 62, 90, 55, 88, 73, 95, 67, 80])

# Calculate statistics
mean_score = np.mean(scores)
median_score = np.median(scores)
std_dev = np.std(scores)

# Grade function
def assign_grade(score):
    if score >= 80:
        return 'A'
    elif score >= 70:
        return 'B'
    elif score >= 60:
        return 'C'
    elif score >= 50:
        return 'D'
    else:
        return 'F'

# Apply grading
grades = [assign_grade(s) for s in scores]

# Output
print("Mean:", mean_score)
print("Grades:", grades)
```

Expected Output:

```
Mean: 77.3
Grades: ['B', 'A', 'C', 'A', 'D', 'A', 'B', 'A', 'C', 'A']
```

6. AI Prompt Journal

Prompt 1:

“Help me build a simple NumPy project for beginners”

- **AI Response Summary:** Suggested project ideas including grade analyzer

- **Usefulness:**
- Helped choose a practical project

Prompt 2:

“Write a full Python code for student grade analyzer using NumPy”

- **AI Response Summary:** Provided full working code
- **Usefulness:**
- Saved development time

Prompt 3:

“Help me fix ModuleNotFoundError: numpy”

- **AI Response Summary:** Guided installation and environment setup
- **Usefulness:**
- Solved environment issue

Prompt 4:

“Guide me to push project to GitHub step-by-step”

- **AI Response Summary:** Provided Git commands and troubleshooting
- **Usefulness:**
- Helped publish project successfully

7. Common Issues & Fixes

Issue 1: NumPy not found

Error:

```
ModuleNotFoundError: No module named 'numpy'
```

Fix:

```
pip install numpy
```

Issue 2: Git not recognized

Error:

```
'git' is not recognized
```

Fix:

- Installed Git from official website
- Enabled “Add to PATH” option

Issue 3: Push rejected

Error:

```
failed to push... fetch first
```

Fix:

```
git pull origin main --allow-unrelated-histories
```

8. References

- NumPy Documentation: https://numpy.org/doc/
- Python Official Docs: https://docs.python.org/
- GitHub Docs: https://docs.github.com/
- YouTube Tutorials (Python & NumPy basics)